Concern Resolution Process

QC Story

QC STORY PROJECT

PROJECT TITLE :- To reduce the Dent and damage defect from Back Door Inside by 70%

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio Body shop.

	Plan	Actual
Start Date	**10/05/12**	**13/05/12**
Completion Date	**30/8/12**	**28/8/12**

Current Status: Completed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

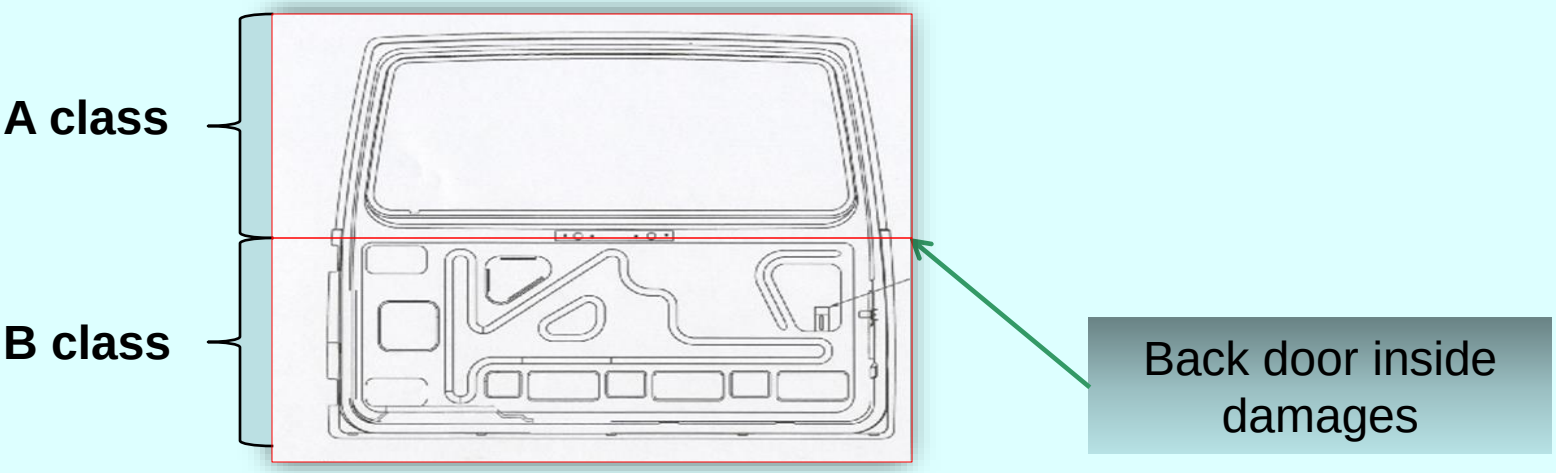
Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Pankaj Yelmame	Deputy Manager	SBS	9922403871	yelmame.pankaj@mahindra.com
2	**Mentor**	Sandeep Deshpande	Manager	MQA (SBS)	9422770889	deshpande.sandeep@mahindra.com
3	**Member 1**	Jagveer Singh	Assit Manager	SBS	9011075653	singh.Jagveer@mahindra.com
4	**Member 2**	Bhat vivek	Deputy Manager	SBS	9922438403	Bhat.vivek @mahindra.com
5	**Member 3**	Gokul Amrutkar	Deputy Manager	MQA (SBS)	9922408133	Amrutkar. Gokul @mahindra.com

1. Select the Theme

Brief description of Problem

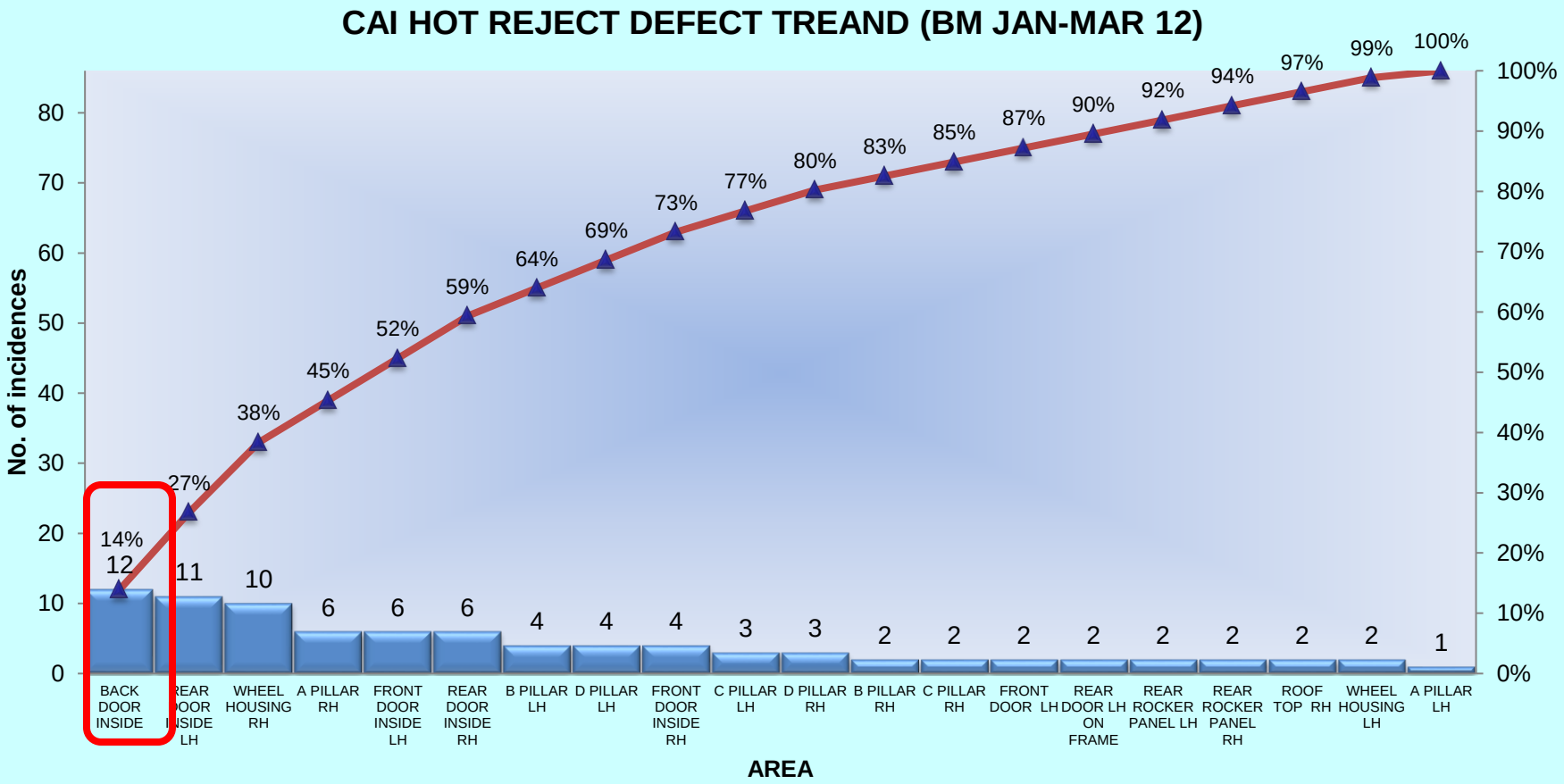
What- To Reduce the Dent and damage defect from back Door Inside.
Where – Major rework (CAI).
When – During WRAP.
Who – Scorpio Body Shop
How – 12 cases observed in Jan-Mar 2012



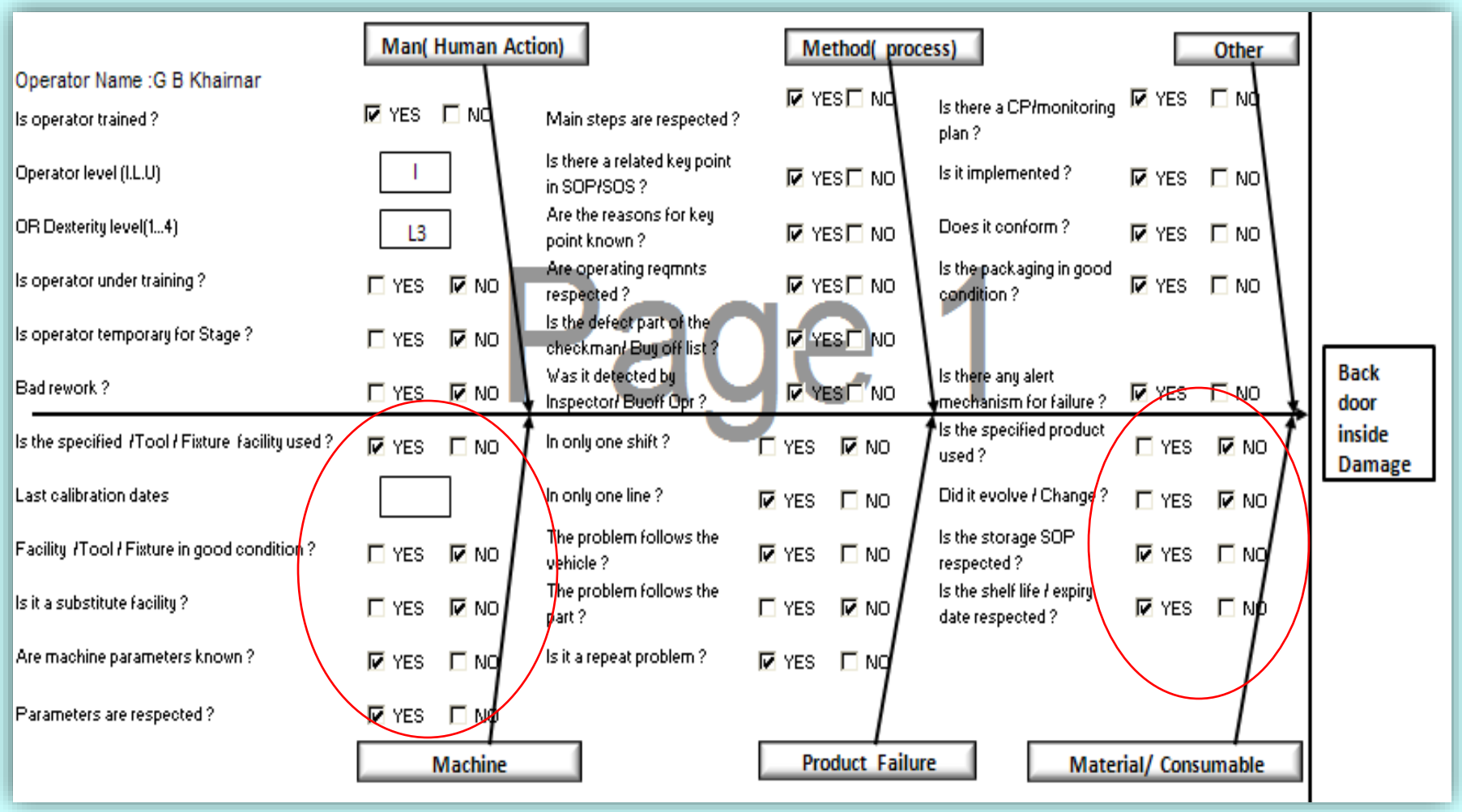
Classwise defect distribution of backdoor inside

2. Justify the choice (reasons)

Pareto for defects of Major rework area for the month of JAN-MAR.20112 justifying the choice



L0 Analysis- Back door damage inside



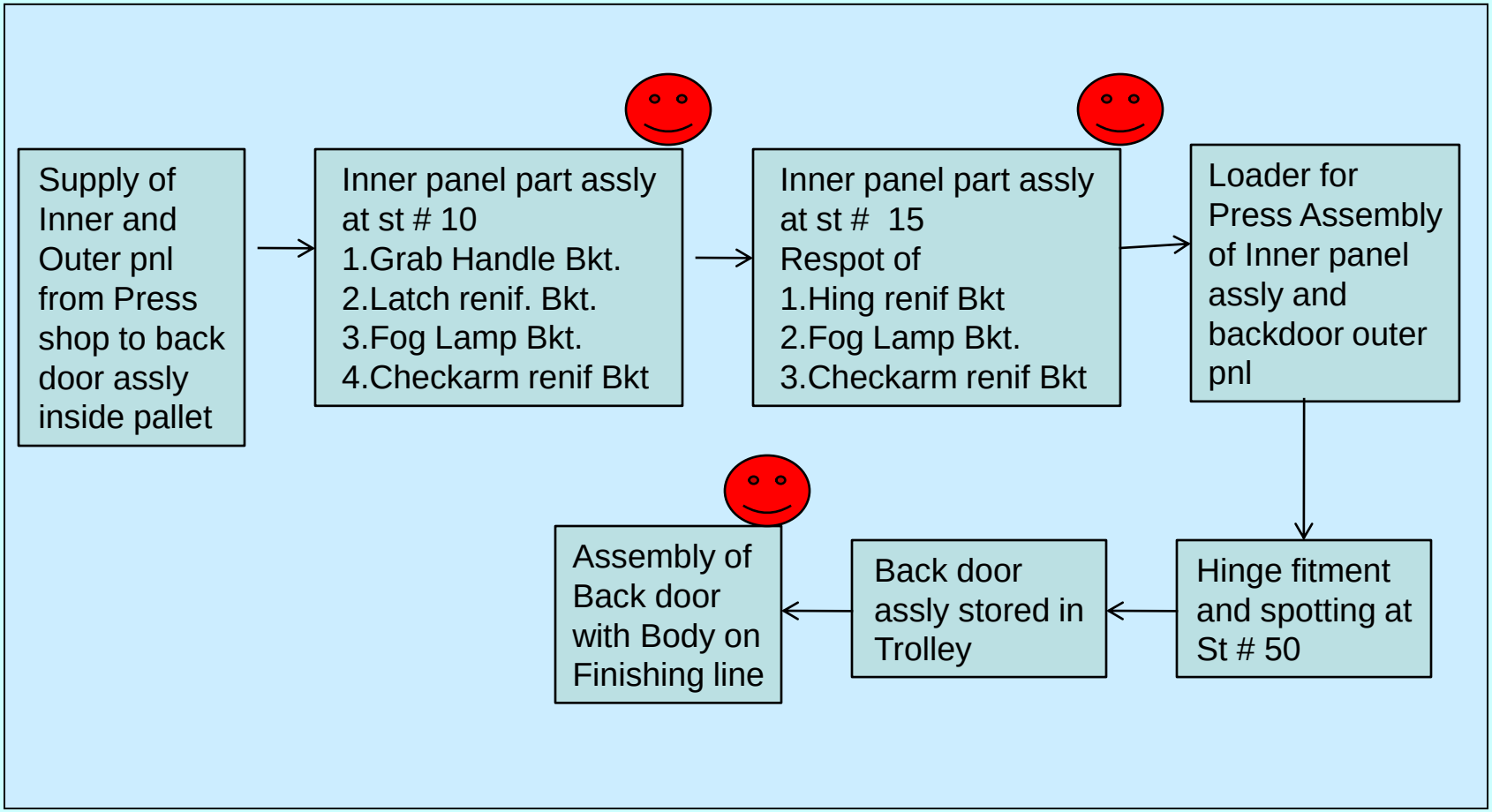
- As per above analysis Machine and Material contributes for defects.

2.b Immediate Containment Action

- Initiated 100 % checking at backdoor assembly.
- Initiated 100 % checking at body buyoff .

3. Understand the current situation

Observation of process



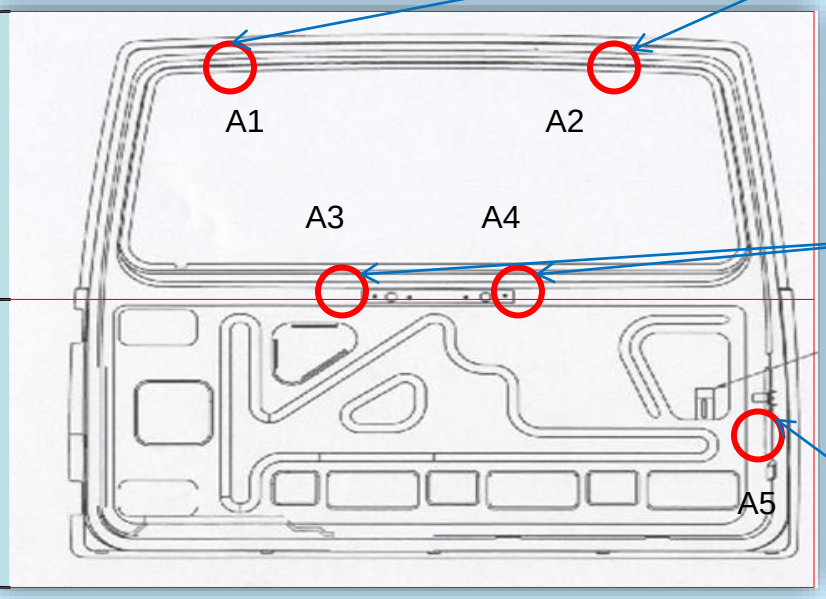
Defect prone area

As per data taken from CAI for Scorpio vehicles, Following diagram shows the class wise bifurcation of damage

From all Backdoor inside damages 12 - V1 (Hot Reject) Concerns found at below mention (A1,A2,A3,A4 & A5) area

A class

B class

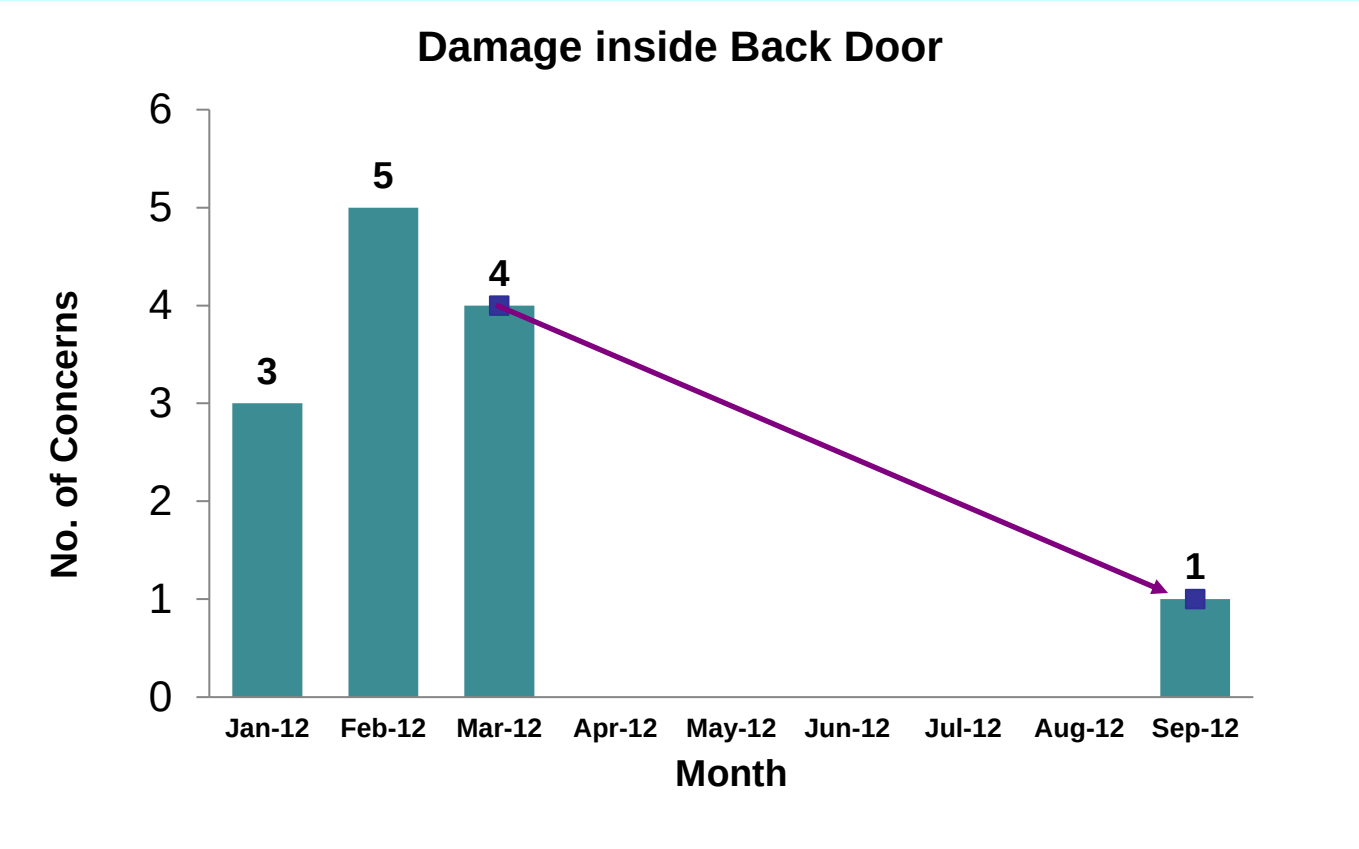


Back door inside Damage Area

4. Select Target

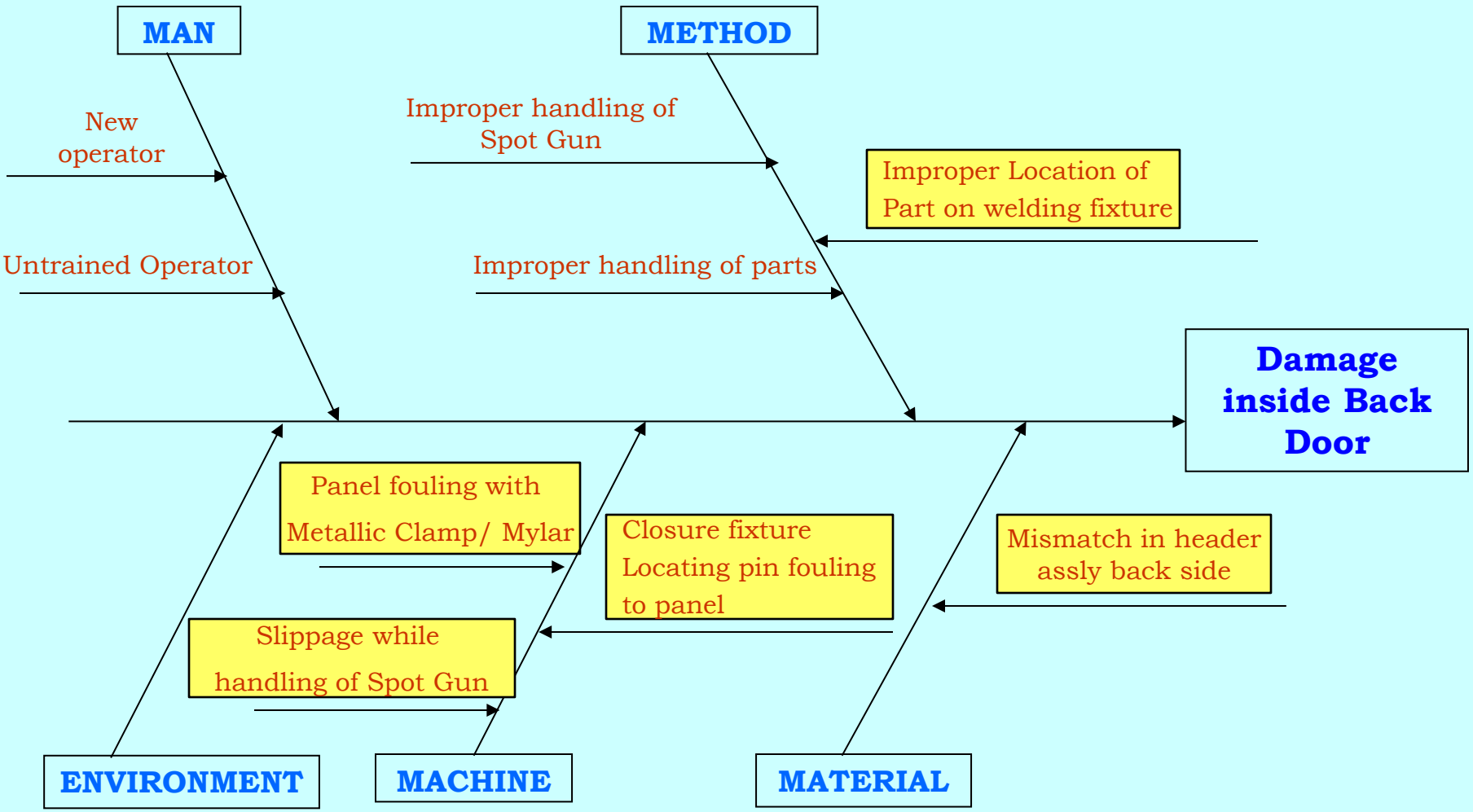
Present Level: Avg. 4 Hot Reject cases reported in Jan to Mar 2012.

Target: Elimination of the concern till Sept 12 by 70 %.



5. Analysis

5.1 Identify possible causes:



List of possible causes

Sr.no.	Possible Causes
1	New operator
2	Untrained operator
3	Slippage while handling of Spot Gun
4	Improper handling of Spot Gun
5	Improper handling of parts
6	Improper Location of Part on welding fixture
7	Panel fouling with Metallic Clamp/ Mylar
8	Closure fixture Locating pin fouling to panel
9	Mismatch in header assly back side fouling to back door

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
1	Slippage while handling of Spot Gun	Spot Gun angle of spotting with panel assembly of 90º checked, also Gun tip position with Cu plate checked in grab handle area	Spot gun got slip at the age of Cu plate at st # 10 after holding Gun 90° with the panel.	valid

Spot welding

No	Probable Cause	Testing method	Observations	Conclusion
2	Improper handling of parts (Inner panel, Latch plate bkt, Grab handle renif bkt, Check arm renif bkt.	Part handling is as per SOS or not	Part handling is as per SOS	Invalid

Part handling is as per SOS

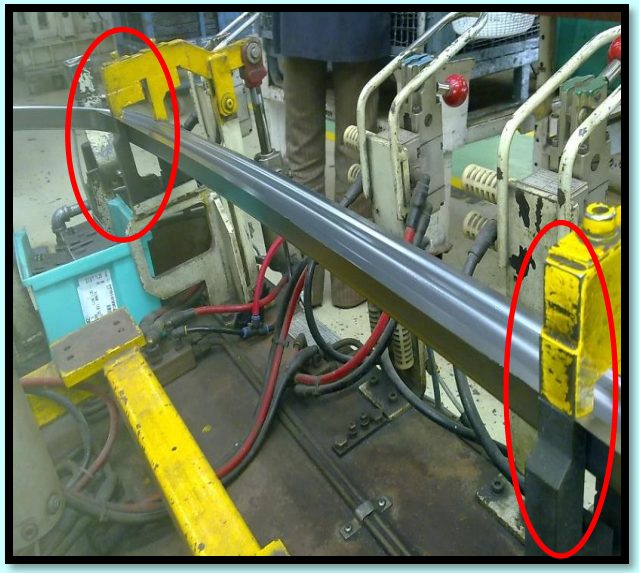
No	Probable Cause	Testing method	Observations	Conclusion
3	Improper Location of Part on welding fixture	As per SOS while loading inner panel assembly on fixture not get foul with metallic locating pin of fixture.	Anomalies found about Inner panel assembly foul with locating pin at st # 20 causing dent at latch area	valid

Fixture locating pin fouling to inner panel at backdoor Assembly stage

No	Probable Cause	Testing method	Observations	Conclusion
4	Panel fouling with Metallic Clamp/ Mylar	All part (Inner panel and inner panel assembly) handling on all fixture checked for fouling with sharp metallic edges of clamp, Mylar and fixture.	All sharp edges from all fixture identified like st # 10 upper Mylar, st # 20 latch area Mylar	valid

Following where sharp edges found causing damages on inner assembly

Sr. no.	Area	No. of Mylar/ Clamp
1	St. # 10 Fixture	3
2	St. # 15 Fixture	3
3	St. # 20 Fixture	4
4	St. # 50 Fixture	5

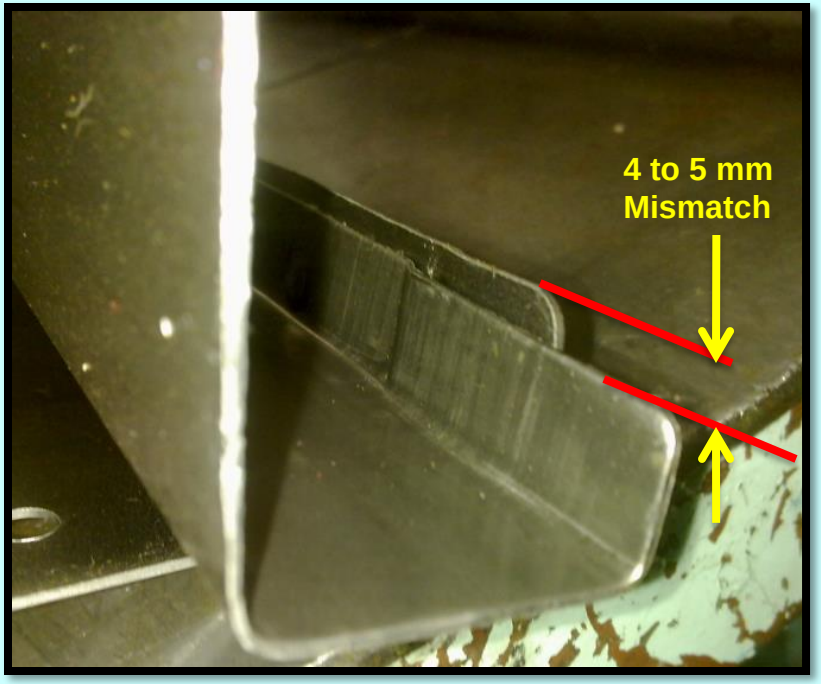


No	Probable Cause	Testing method	Observations	Conclusion
5	Closure fixture Locating pin fouling to panel	Visual inspection- Closure fixture manually operated overhang type, front portion of locating pin foul on inner panel easily	Front portion of pin foul and create dent mark on inner panel	Valid

Fixture locating pin fouling to inner panel at backdoor fitment stage closure area

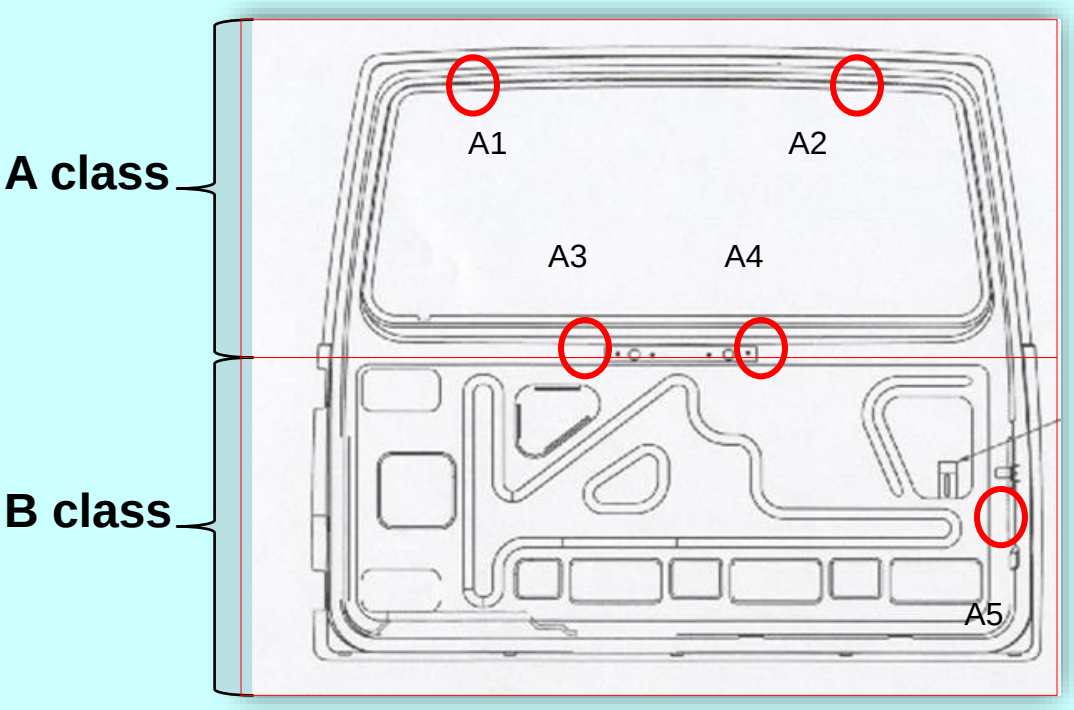
No	Probable Cause	Testing method	Observations	Conclusion
6	Mismatch in header assly back side fouling to back door	Header assembly both inner and outer panel mismatch measure 4 to 5 mm found	Panel mismatch assembly received from supplier (MUSCO)	Valid

Body assembly rear header both panel mismatch found



CAI Hot reject report of back door inside

Back door inside Damage Area



Location	Area	No. of HR Defect
A1	Above inside glass opening area LH	2
A2	Above inside glass opening area RH	2
A3	Inside grab handle area LH	1
A4	Inside grab handle area RH	3
A5	Inside Near latch area	4

WHY - WHY Analysis

5.3 Identify the root causes: (Use Format MQS/MAPS/F05)

Valid Probable Cause	WHY	WHY	WHY	WHY	WHY
Slippage while handling of Spot welding Gun	**Blind operation while spotting in grab handle area**	**Short width of Cu plate**	**Improper CU plate design**		
Improper Location of Part on welding fixture	**Locating pin not visible while keeping the part on fixture**	**Operator holding the panel front of him**	**Assembly Fixture is at the back side of panel**		
Panel fouling with Metallic Clamp/ Mylar	**Sharp edges of Mylar**	**Close tolerances bet panel and fix**	**Fixture design constraint**		
Closure fixture Locating pin fouling to panel	**Pin striking to inner panel while locating assly in fixture by operator**	**Inadequate \Blind positioning during the fitment process**			
Mismatch in header assly back side fouling to back door	**Rear header not as per specification**	**Inadequate constraint of assy in fixture at vendor end**			

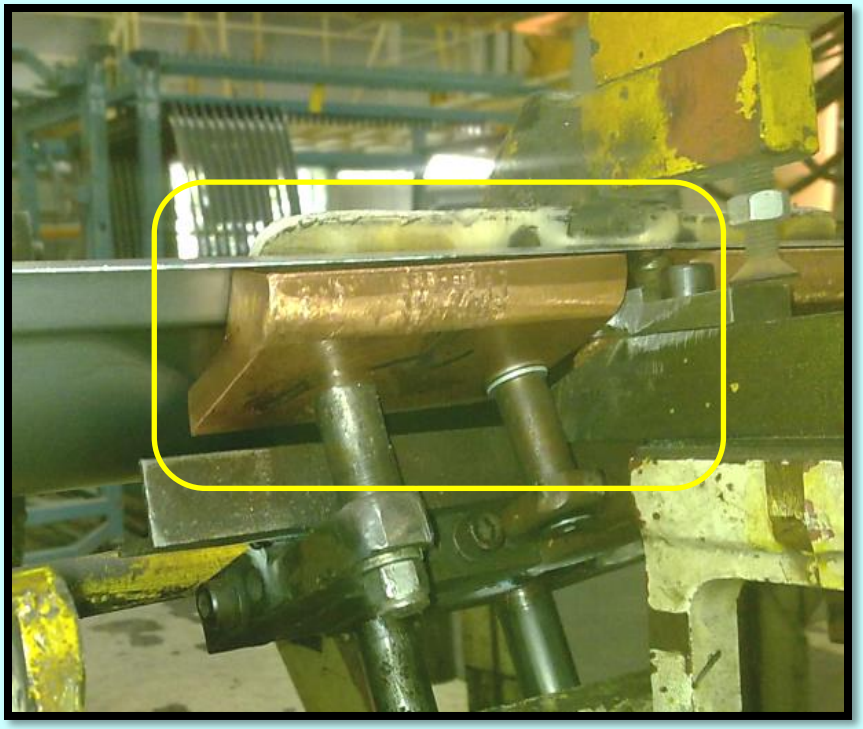
6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Slippage while handling of Spot Gun	Profile shape CU plate on both side of Grab handle area fitted to avoid slippage of gun and spot damage	20/5/12	Pankaj Vivek Manoj	completed
Improper Location of Part on welding fixture	Fixture with separate tilter provided on st # 20 to avoid fouling with pin	25/5/12	Vivek Pankaj	completed
Panel fouling with Metallic Clamp/ Mylar	Fixture Mylar Design change by providing fillet on edge instead of sharp edges	5/6/12	Pankaj Gajanan Jagveer	completed
Closure fixture Locating pin fouling to panel	Rubber protected parallel supportive rod provided which foul prior to metallic pin	13/6/12	Pankaj Vivek	completed
Mismatch in header assly back side fouling to back door	Body Rear header assembly corrected at supplier (MUSCO) end as per design	16/8/12	Pankaj Gokul Pravin	completed

6. Implement the corrective measures

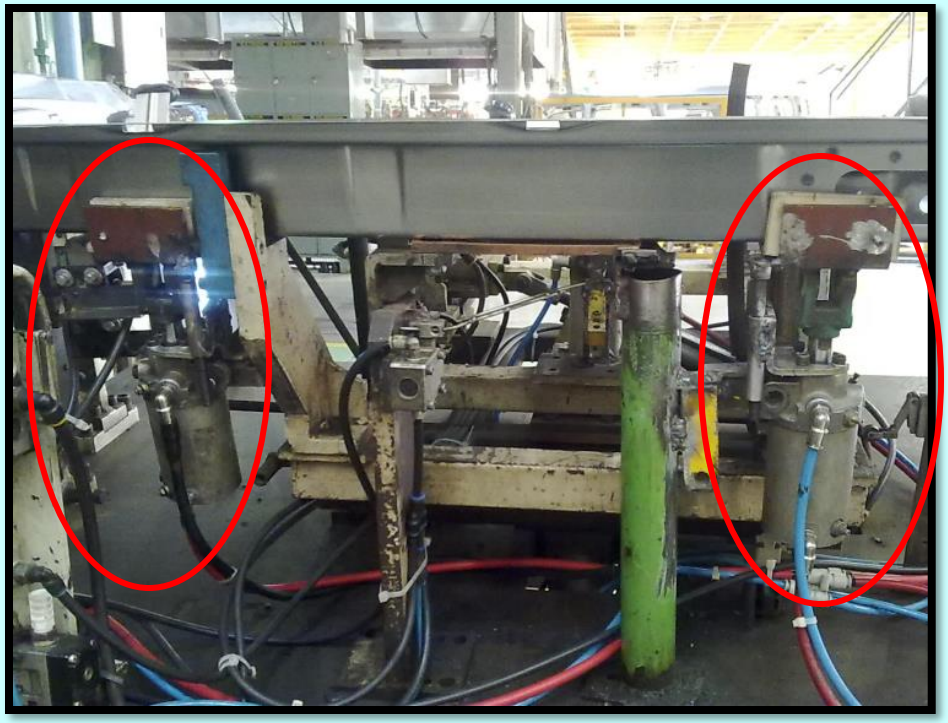
Straight edge type cu plate at st # 10 back door assembly



Back door Inner panel profile shape CU plate fitted on St # 10

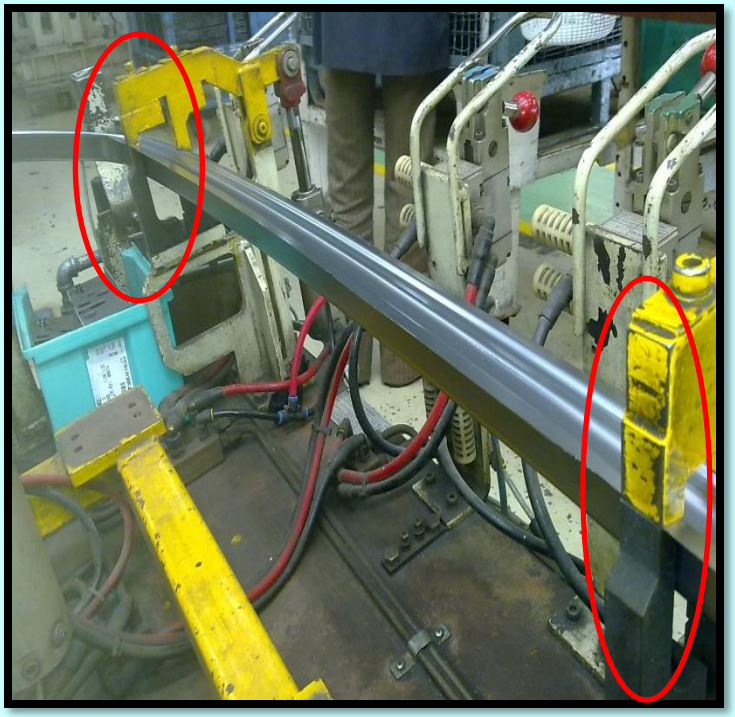
6. Implement the corrective measures

Fixture locating pin fouling to inner panel at backdoor Assembly stage



Fixture with separate tilter provided on st # 20 to avoid fouling with pin

6. Implement the corrective measures



Fixture clamp/ Mylar fouling to inner panel at backdoor Assembly stage

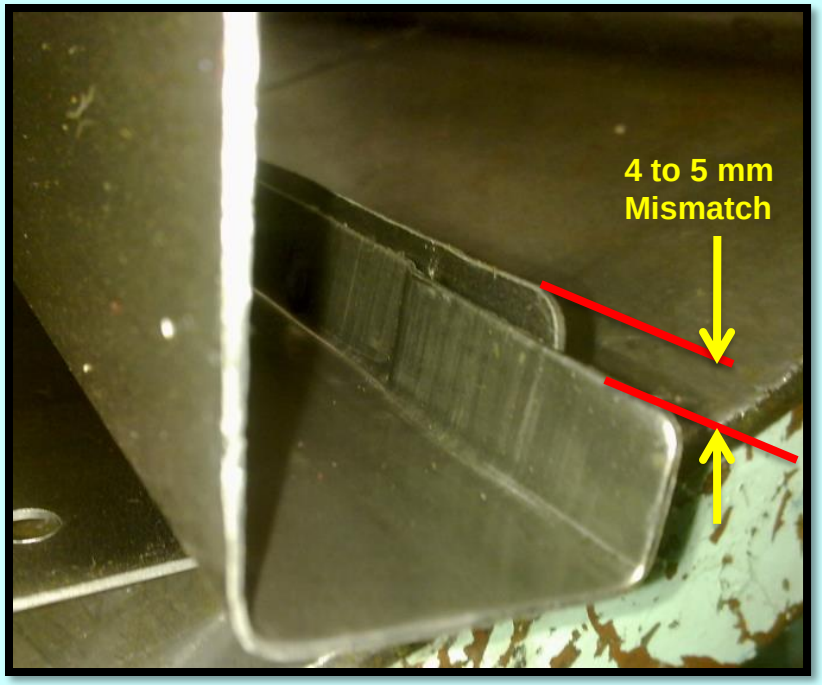
Fixture Mylar Design change by providing fillet on edge instead of sharp edges

6. Implement the corrective measures

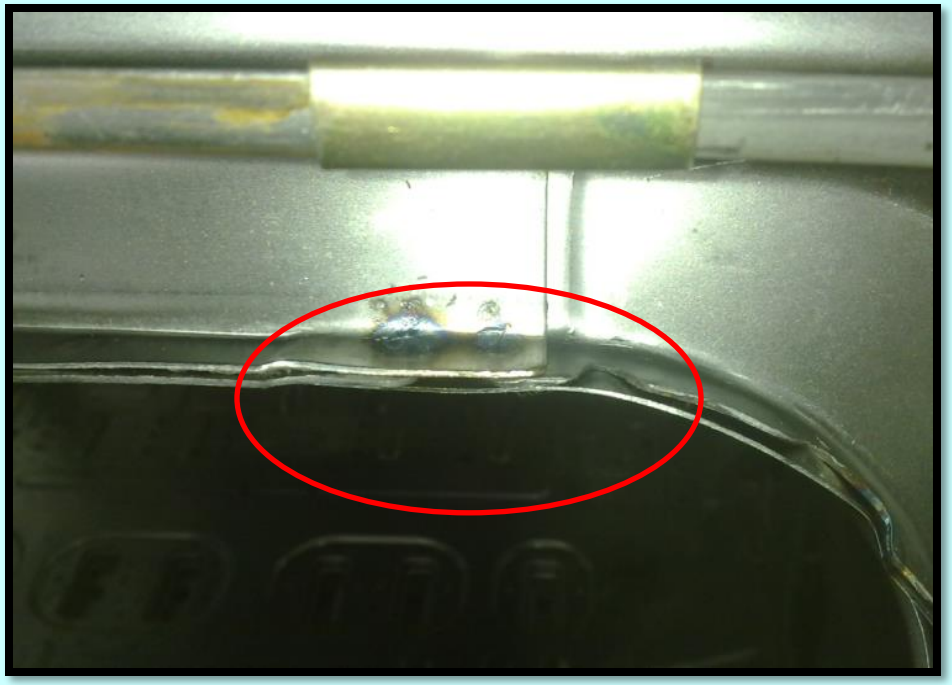
Fixture locating pin fouling to inner panel at backdoor fitment stage closure area

Rubber protected parallel supportive rod provided which foul prior to metallic pin

6. Implement the corrective measures



Body assembly rear header both panel mismatch found

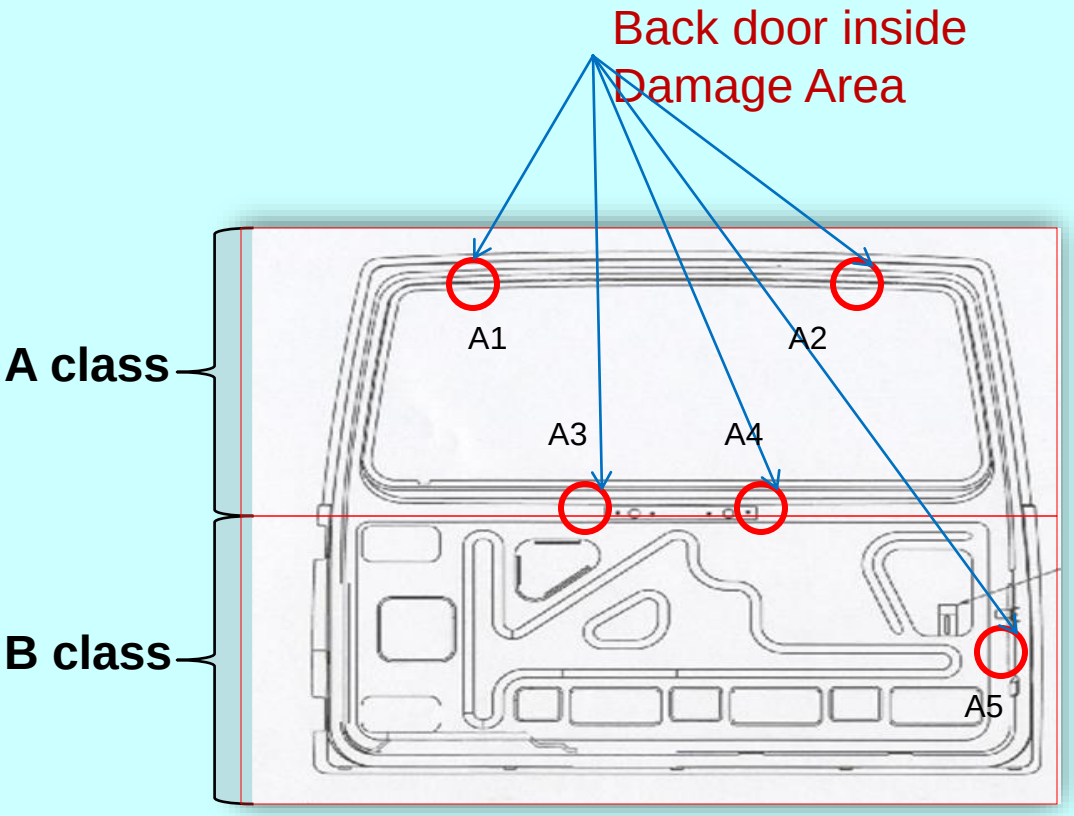


Body Rear header assembly corrected at supplier (MUSCO) end as per design

7. Check & confirm the results & effects

7.1 Verify Results:

After implementation of corrective measures at supplier end, and inside body shop it was observed Damages are reduced.
Following were the readings of damages area wise

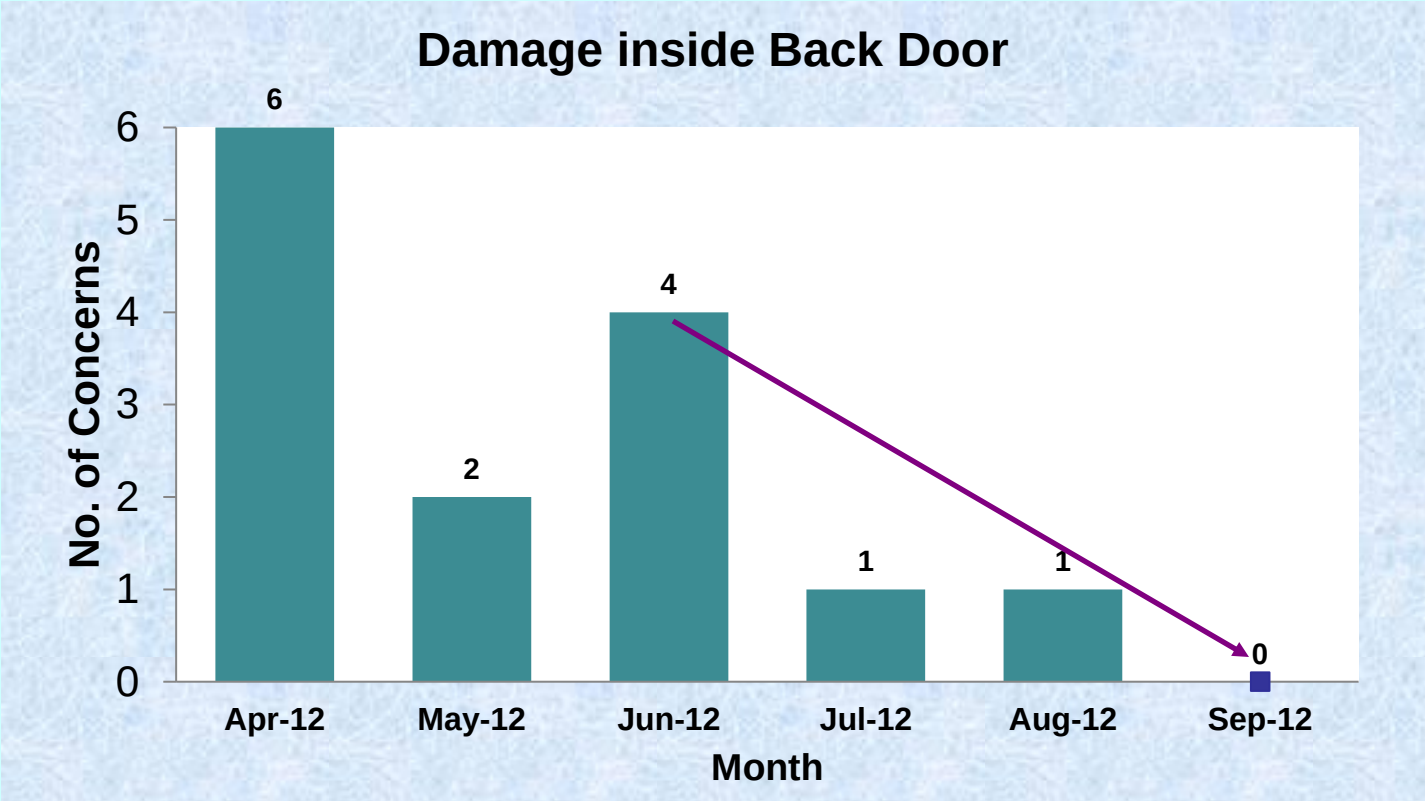


Location.	Apr 12	May 12	Jun 12	Jul 12	Aug 12	Sep 12 Till
A1	2	0	0	0	0	0
A2	0	1	2	1	0	0
A3	1	1	0	0	0	0
A4	1	0	1	0	0	0
A5	2	0	1	0	1	0

7. Check & confirm the results & effects

7.1 Verify Results:

After implementation of corrective measures at supplier end, the concern of back door inside damage reduced in the month of Sept 2012 so far as reflected in trend shown below



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

Change implemented 100% from 16 August 2012.

7.3 Tangible & Intangible benefits:

1. **STR improvement at CAI. (0.5%)**
2. **Reduction in rework cost. (Appx. Rs 50,000/- per anum)**
3. **Customer satisfaction.**

8. Standardisation

8.1 Updation of documents:

SOS updated,

Fixture history card :

(Fixture drawing no. 0103CG0140N (F1-00-00) ,306-WF-000,107000016020)

Fixture validation record attached : 6 June 2012

8.2 Training:

Operators trained at Back door assly for handling of panel and Spot welding Gun, immediate damage detection at the final stage.

8.3 Additional Audit / Check

1. Daily monitoring through Supervisory Check Sheet.

9. Summarise & Plan future actions

9.1 Conclusion:

Failure mode to be analyze prior to design of fixture.

9.2 Horizontal Deployment:

Various fixture improvement to be implemented for other side door assembly

9.3 Lessons Learnt:

Improper fixture and panel handling leads to create damages on part

Thank You

Fixture history card

FIXTURE HISTORY CARD					
ZONE:	Z20				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO.	Z20-10		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	Assy Back door 10				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
1/10/2010			Damage check arm flange	check arm flange guide extended	QCRT CA3P0075/10
6/11/2010			Back Door Latch Area Spot Damage	Guard provided to avoid spot on cu plate edge	
21/12/10			Damage back door Grab Handle	Cu plate float restriction by providing support to the plate opposite to the Pin	QCRT CA3P0081/10
15/1/11			Dent-Inner panel	Gap created in panel and myler in asthetic area by removing shims and providing additional resting at the flange.	
18/05/11			Heavy mismatch latch area	Additional stopper provided to arrest the latch brkt movement	Scrap-May-11
10/7/2011			Heavy damage on latch hole	Copper plug provided	Scrap July-11
24/07/2011			Back door latch area pin hole	New copper plate provided	CA2P0416
16/12/2011			Damage check arm mtg area	Stopper provided	CA2P0495
16/01/12			Spot twist on latch area	Stopper provided to avoid spot on edge	CA2P00508
24/1/12			Back door damage near Grab handle	Profile cu plate provided of Grab handle	i4-68829
16/03/12			Damage back door latch area	V block groove \relief provided	CA2P0538
25/3/12			Damage in Check arm area inside back door	Provided interlocking in Check arm clamp	i4-64481
24/4/12			Damage at check area while spotting	Provided cu plate guard instead of Nylon	i4-85474
20/5/12			Spot damage upper grab handle	Provided fillet to profile shape cu plate at grab handle	
20/5/12			Damage at check area while spotting	Nylon guard provided for check arm Bkt	i4-91539
12/6/2012			Inner lower portion of check arm spot hole	Provided cu plate guard	

Page 1

Fixture history card

FIXTURE HISTORY CARD					
ZONE:	Z20				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO.	Z20-15		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	Assy Back door 15				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	
8/8/08			Damage/Spark on inner assy because of gun hit	Nylon gun guard provided instead of Spring steel guard.	
26/7/09			Dent on back door inner panel.	Guide rod provoided to bottom side inner panel resting	QCRT CA 3P 0071/10 (i4-201638)
9/9/2010			Back Door inside damage near Hinge mounting area	Hinge side SS Guard provided with ED coating.	Scrap Sept -10/CA3P71(RO)
12/11/2010			Spot Damage - Backdoor Fog lamp spots	Stopper provided to Cu plate to restrict its movemnt in Z direction whie spotting.	QCRT CA2P323/10
20/9/10			Dent - inner panel Backdoor	Touching myler chamfered amd sharp edges removed.	
22/4/2011			Back door inner panel damage	Pipe with covered rubber hose more ht than locating pin provided.	i4-321268
5/7/2011			Back door inner panel damage	Intermediate 3 cylinder mounted on Fixture.	
5/26/2011			Dent back door inside bottom	Single piece spring steel guard	CA3P098
25/12/11			Damage at back door inner at bottom	Provided nylon guard from gun hitting	i4-10152
23/1/12			Clinching open due to flange damage	MS Gun restrictor provided on clamp	
18/2/12			Damage near fog clamp area	Provided nylon guard from gun hitting	i4-Provided nylon guard
12/3/2012			Back door hinge fouling to panel	Locating pins shifted by 1mm to make reinf. In central position	CA2P528
25/5/12			Back door inner panel damage	cylinders with nylon pad mounted on Fixture.	

Page 1

Fixture validation record

FARO CONTROL STATION

OPERATOR :NITIN
PART NAME : Back Door Assy
Z20-15
DATE: June 06 2012

FARO CAM2

M_4300X *(Measured Comp Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	4299.6	4299.84	0.5	-0.5	-0.24	*******
	Y	-10.03	-10	0.5	-0.5	-0.03	*******
Center	Z	1949.29	1949.44	0.5	-0.5	-0.15	*******

M_4306X *(Measured Comp Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	4306.83	4306.7	0.5	-0.5	0.13	*******
	Y	10.04	10	0.5	-0.5	0.04	*******
Center	Z	1936.15	1936.11	0.5	-0.5	0.04	*******

M_4303X *(Measured Comp Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	4303.91	4303.82	0.5	-0.5	0.09	*******
	Y	-10.2	-10	0.5	-0.5	-0.2	*******
Center	Z	1931.72	1931.24	0.5	-0.5	0.48	*******